PROBABILITY AND STATISTICS FOR NON-PATTERN PROBLEMS

CHAPTER 7

STATISTICAL HYPOTHESES TESTING

CHAPTER OUTLINE

- Define a hypothesis and hypothesis testing.
- Describe the hypothesis-testing procedure.
- A one-tailed and a two-tailed test of hypothesis.
- Test of hypothesis about a population mean.
- Test of hypothesis about a population proportion.
- Define Type I and Type II errors
- Compute the probability of a type II error.

WHAT IS HYPOTHESIS TESTING?

- A hypothesis is **a statement about the value of a population parameter** developed for the purpose of testing.
- **Example** of hypotheses made about a population parameter are:
  - The student watch an average of 3 hours social media per day.

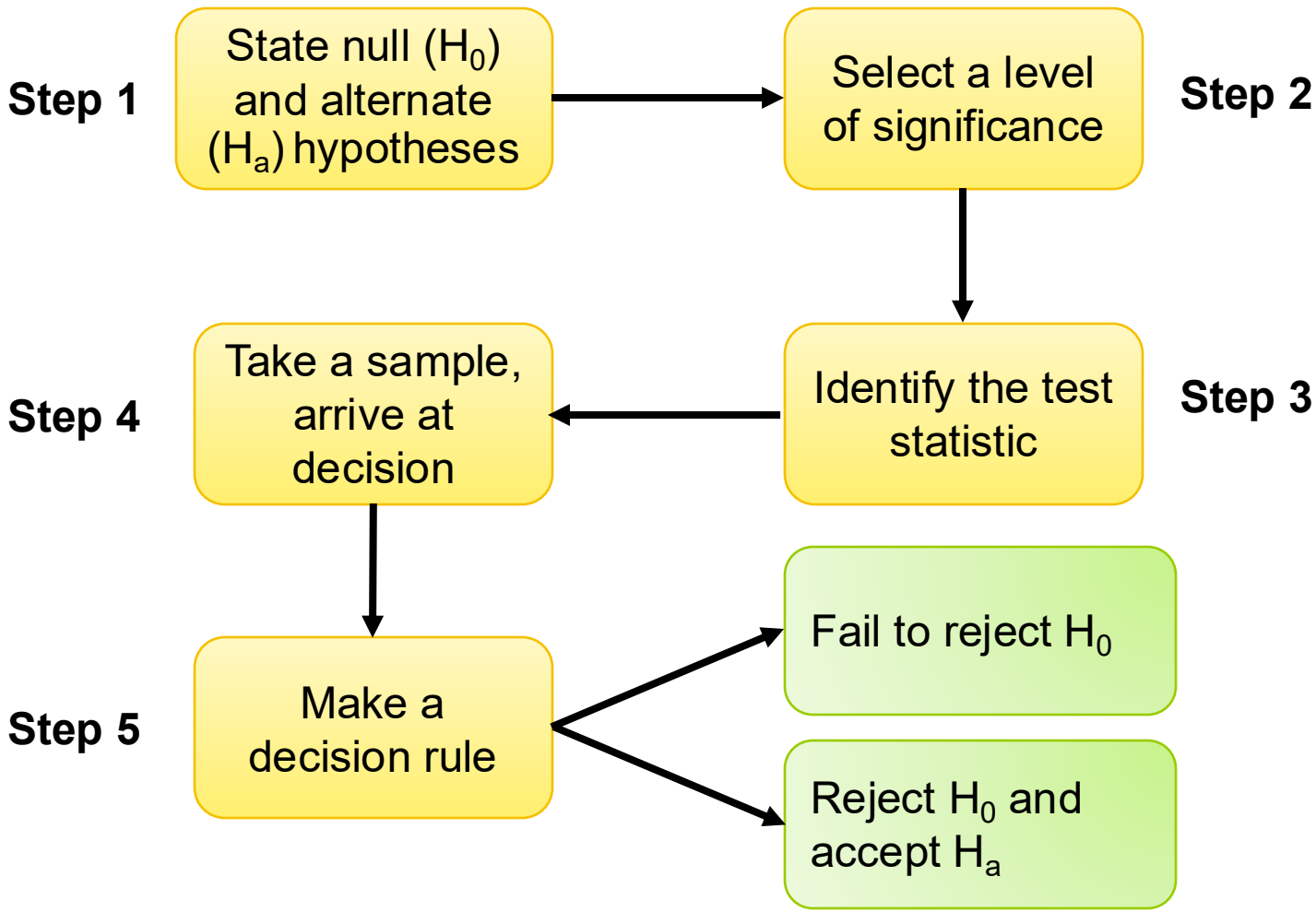
WHAT IS HYPOTHESIS TESTING?

- **Hypothesis testing** is a procedure, based on sample evidence and probability theory, used to determine whether the hypothesis is a reasonable statement and **should not be rejected**, or is unreasonable and **should be rejected.**

The statements are called hypotheses.
The decision-making procedure is called hypothesis testing.

Hypothesis testing involves taking a random sample,
computing a **test statistic** from the sample data,
and then using the test statistic to **make a decision about the null hypothesis.**

HYPOTHESIS TESTING PROCEDURE



IMPORTANT THINGS TO REMEMBER ABOUT H_0 AND H_a

- H_0: null hypothesis, H_a: alternate hypothesis
- H_0 and H_a are **mutually exclusive** and **collectively exhaustive**
- H_0 is always presumed to be true
- H_a has the burden of proof
- A random sample (n) is used to "**reject H_0**"

IMPORTANT THINGS TO REMEMBER ABOUT H_0 AND H_a

- If we conclude "do not reject H_0", this does not necessarily mean that the H_0 is true, it only suggests that there is not sufficient evidence to reject H_0.
- Rejecting the H_0 then, suggests that the H_a **may** be true.
- Equality or boundary is always part of H_0 ("=", "$\geq$", "$\leq$").
- "$\neq$", "<" and ">" always part of H_a

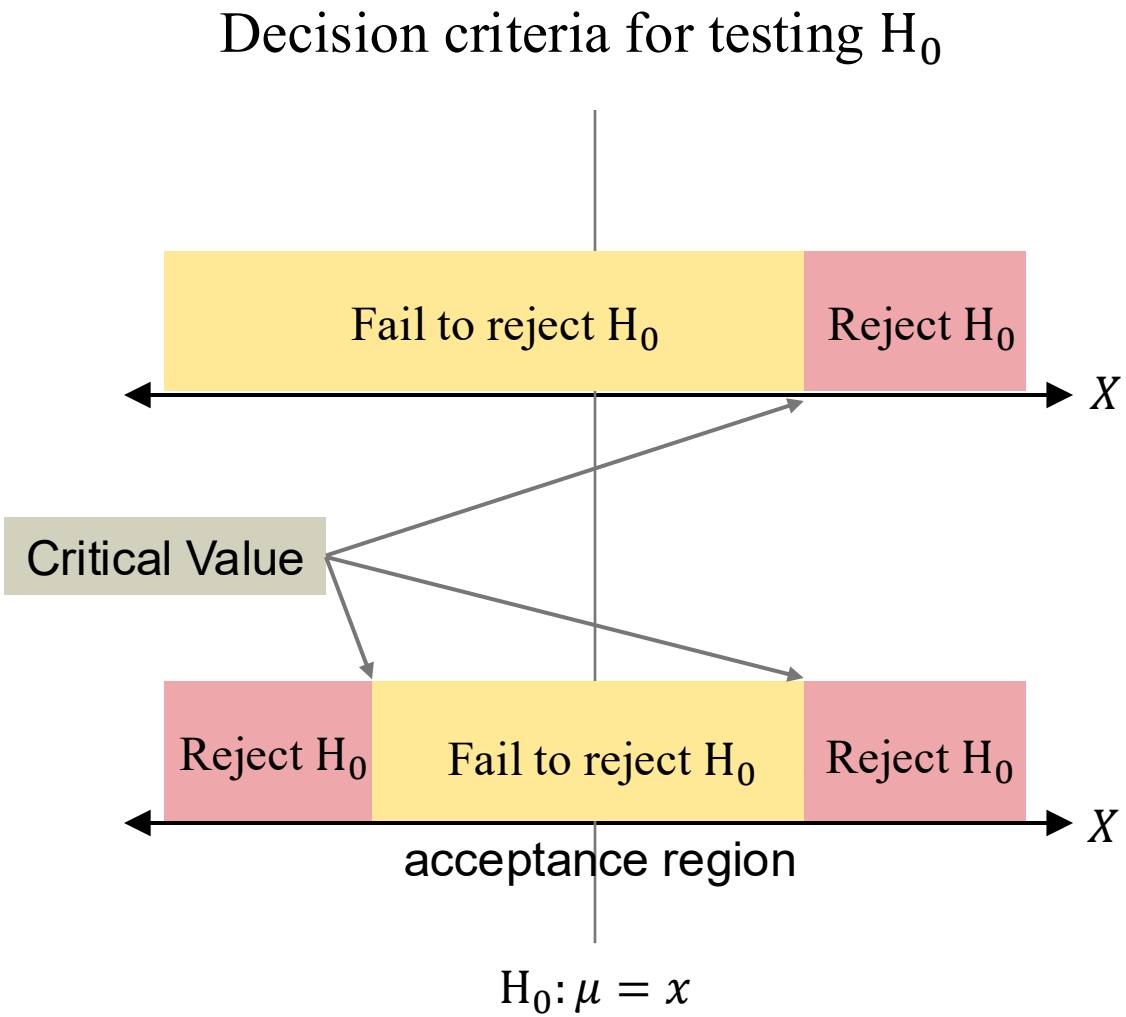
ONE-SIDED, TWO-SIDED ALTERNATIVE HYPOTHESIS

$H_0: \mu = x$

$H_a: \mu > x$

$H_0: \mu = x$

$H_a: \mu \neq x$



HOW TO SET UP A CLAIM AS HYPOTHESIS

- In actual practice, the **status quo** is set up as H_0
- If the claim is "boastful" set up as H_a.
- In problem solving, look for **key words** and convert them into symbols.
  - Some key words: "*improved, better than, as effective as, different from, has changed*, etc."

LEFT-TAIL OR RIGHT-TAIL TEST?

- **A left-tailed test** is used when the H_a states that the true value of the parameter specified in the H_0 is **less than** the H_0 claims.

- **A right-tailed test** is used when the H_a states that the true value of the parameter specified in the H_0 is **greater than** the H_0 claims.

LEFT-TAIL OR RIGHT-TAIL TEST?

- The direction of the test involving claims that use the words "has improved", "is better than", will depend upon the variable being measure.
  - Ex., If the variable involves **time** for a medication to take effect, the words "*better*" "*improve*" or "*more effective*" are translated as "<" (less than), i.e., faster relief.
  - OTOH, if the variable refers to a test score, then the words *"better" "improve"* or *"more effective"* are translated as ">" (greater than) i.e., higher test scores

LEFT-TAIL OR RIGHT-TAIL TEST?

Keywords	Symbol	Part of:
Larger (or more) than	$>$	H_a
Smaller (or less)	$<$	H_a
No more than	$\leq$	H_0
At least	$\geq$	H_0
Has increased	$>$	H_a
Is there difference?	$\neq$	H_a
Has not changed	$=$	H_0
Has “improved”, “	$>$	H_a

CRITERIA FOR DECISION?

Significance Level (α):

- The **probability of rejecting** the null hypothesis when it is true. Normally referred in percentage

The level of Confidence :

- Probability of confidence

LEVEL OF SIGNIFICANCE

- Accept the Null Hypothesis if the sample mean **is close** to the mean in the Hypothesis.
- The significance level, denoted as alpha: α, determines how far out from the null hypothesis value.
- To graph a significance level of 0.05, we need to shade the 5% of the distribution that is furthest away from the null hypothesis.

LEVEL OF SIGNIFICANCE

- **One tailed test**

Right tail test

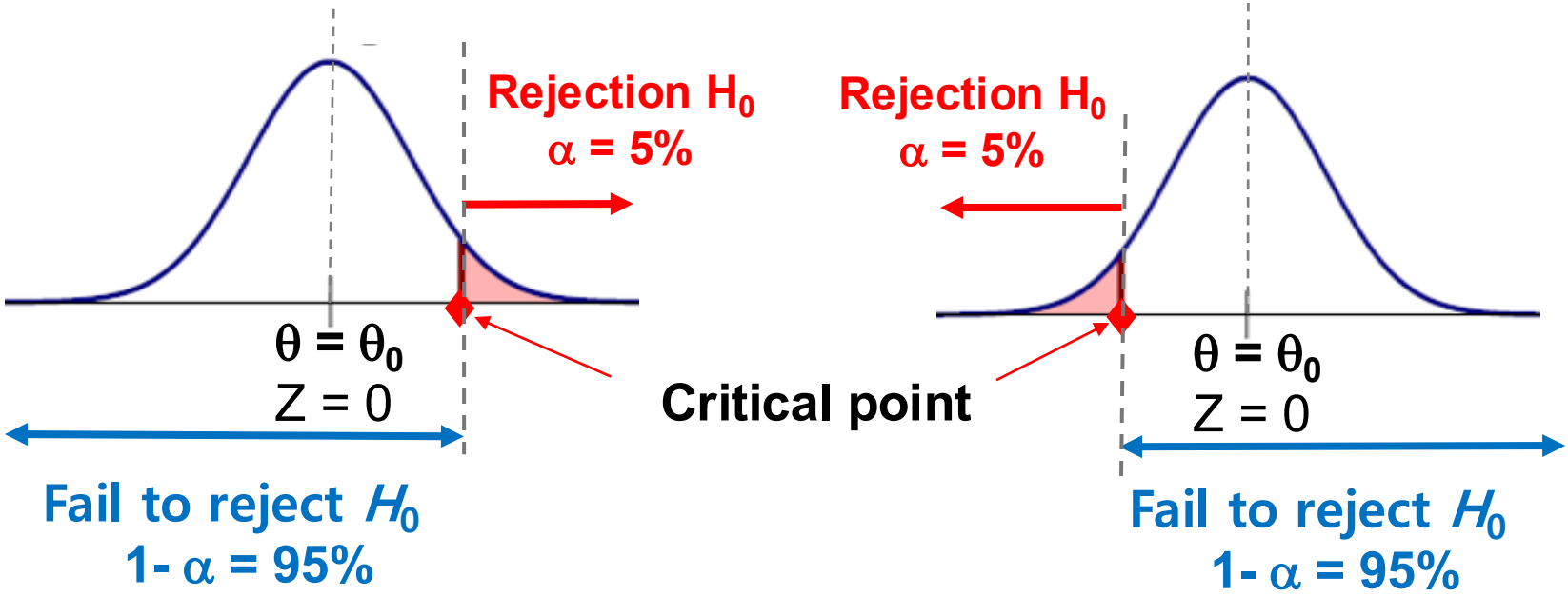
$H_0 : \theta \leq \theta_0$

$H_a : \theta > \theta_0$

Left tail test

$H_0 : \theta \geq \theta_0$

$H_a : \theta < \theta_0$

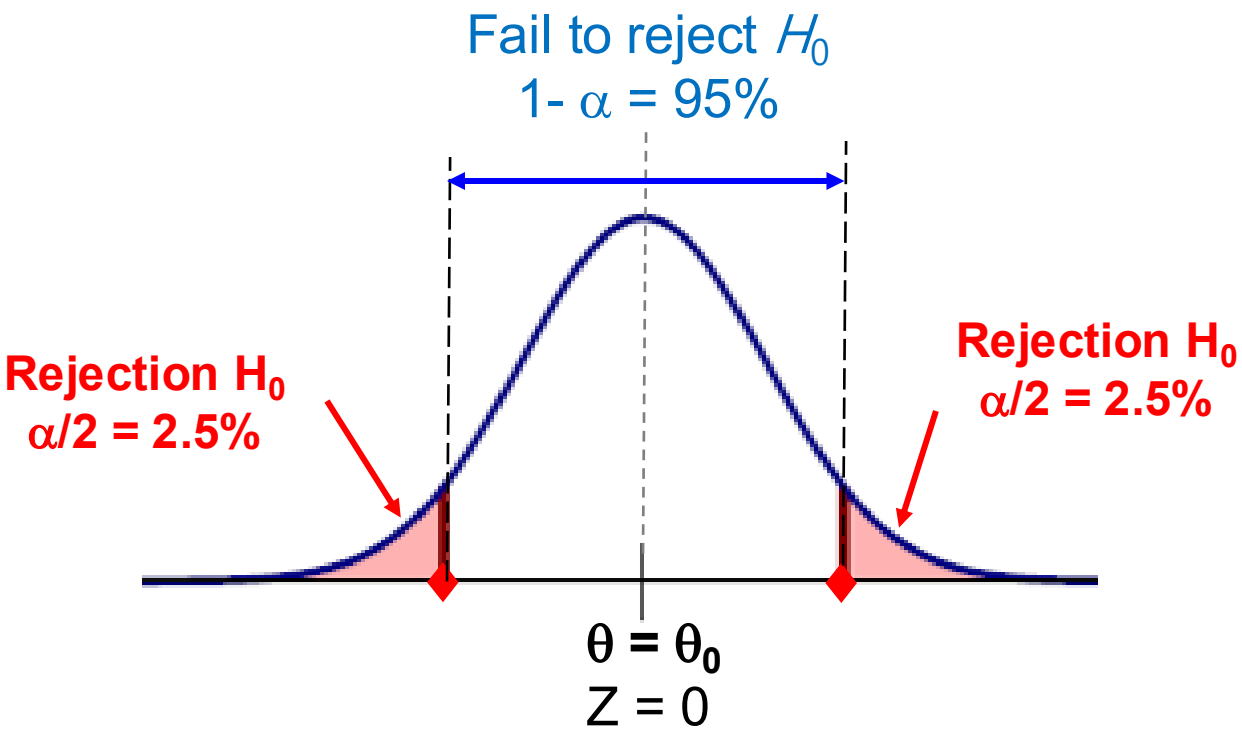


LEVEL OF SIGNIFICANCE

- **Two tailed test**

$H_0 : \theta = \theta_0$

$H_a : \theta \neq \theta_0$



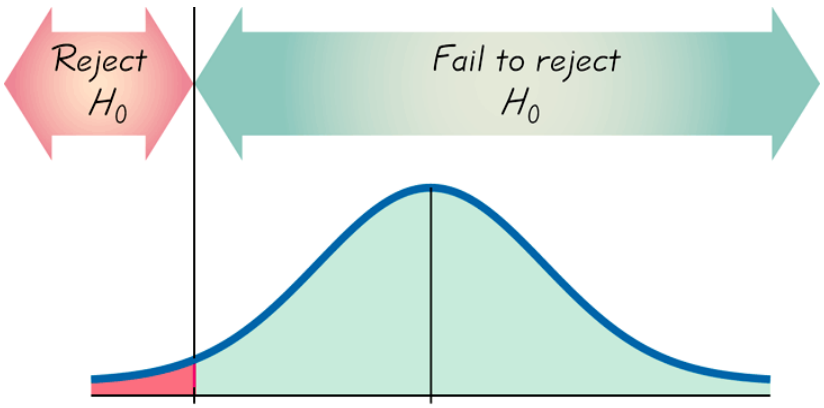
CRITICAL REGION

- **Critical Region**

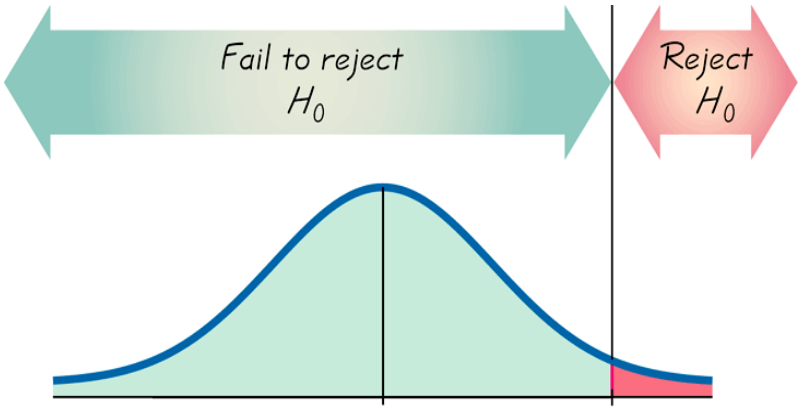
The **critical region** or **rejection region** is the set of all values of the test statistic that cause us to reject the null hypothesis.

Ex. the red-shaded region.

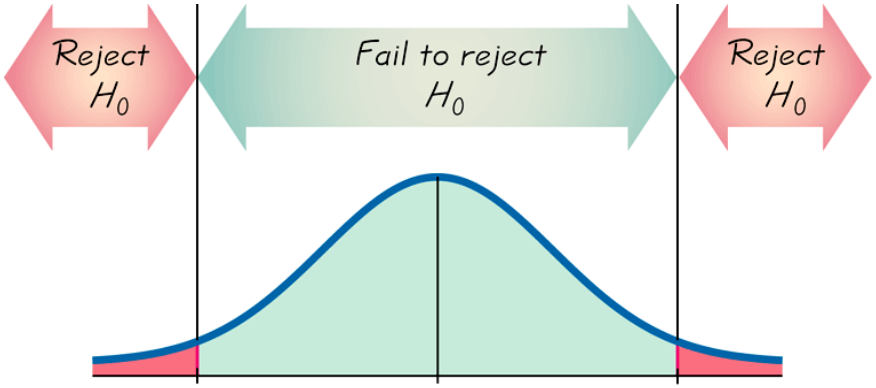
CRITICAL REGION



Sign used in H_1: $<$



Sign used in H_1: $>$



Sign used in H_1: $\neq$

HYPOTHESIS TESTS FOR THE MEAN

Hypothesis Tests for μ

σ Known (Z-test)

$$Z = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

σ Unknown (*t*-test)

$$t = \frac{\bar{X} - \mu_0}{\frac{S}{\sqrt{n}}}$$

Z-Statistic, Z-score, Standard score

HYPOTHESIS SETUPS FOR TESTING A MEAN (μ)

- **Two-tailed Test**

$$H_0 : \mu = \mu_0$$

$$H_a : \mu \neq \mu_0$$

Reject H_0 if :

$$|Z| > Z_{\alpha/2}$$

$$Z = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

$$|t| > t_{\alpha/2}$$

$$t = \frac{\bar{X} - \mu_0}{\frac{S}{\sqrt{n}}}$$

HYPOTHESIS SETUPS FOR TESTING A MEAN (μ)

Left-tailed Test

$$H_0 : \mu \geq \mu_0$$

$$H_a : \mu < \mu_0$$

Reject H_0 if :

$$Z < -Z_\alpha$$

$$t < -t_{\alpha,\ d.f.=n-1}$$

$$Z = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

Right-tailed Test

$$H_0 : \mu \leq \mu_0$$

$$Ha : \mu > \mu_0$$

Reject H_0 if :

$$Z > Z_\alpha$$

$$t > t_{\alpha,\ d.f.=n-1}$$

$$t = \frac{\bar{X} - \mu_0}{\frac{S}{\sqrt{n}}}$$

HYPOTHESIS SETUPS FOR TESTING A PROPORTION (p)

- Applicable to situations that asking what percentage of the population is in favor of a certain idea or concept such as:
  - Will a new product be a success or not?
  - Will this certain candidate win or not?

HYPOTHESIS TESTING FOR A PROPORTION p

- **Two-tailed Test**

$$\mathrm{H}_0 : p = p_0 \qquad \text{Reject } \mathrm{H}_0 \text{ if : } |Z| > Z_\alpha$$

$$\mathrm{H}_a : p \neq p_0$$

- **Left -tailed Test**

$$\mathrm{H}_0 : p \geq p_0 \qquad \text{Reject } \mathrm{H}_0 \text{ if : } Z < Z_\alpha$$

$$\mathrm{H}_a : p < p_0$$

- **Right -tailed Test**

$$\mathrm{H}_0 : p \leq p_0 \qquad \text{Reject } \mathrm{H}_0 \text{ if : } Z > Z_\alpha$$

$$\mathrm{H}_a : p > p_0$$

Z TEST FOR PROPORTION

A statistical test for population proportion. The Z test can be used on the assumption that $\boldsymbol{np_0 \geq 5}$ and $\boldsymbol{n(1 - p_0) \geq 5}$

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}}, \qquad \hat{p} = \frac{x}{n}$$

- $\hat{p}$ = sample proportion
- p_0 = population proportion
- n = sample size
- x = number of successes in sample

EX. HYPOTHESIS TESTING FOR A POPULATION MEAN WITH KNOWN σ

An article states that student watch an average of 3 hours social media per day. Is this true?

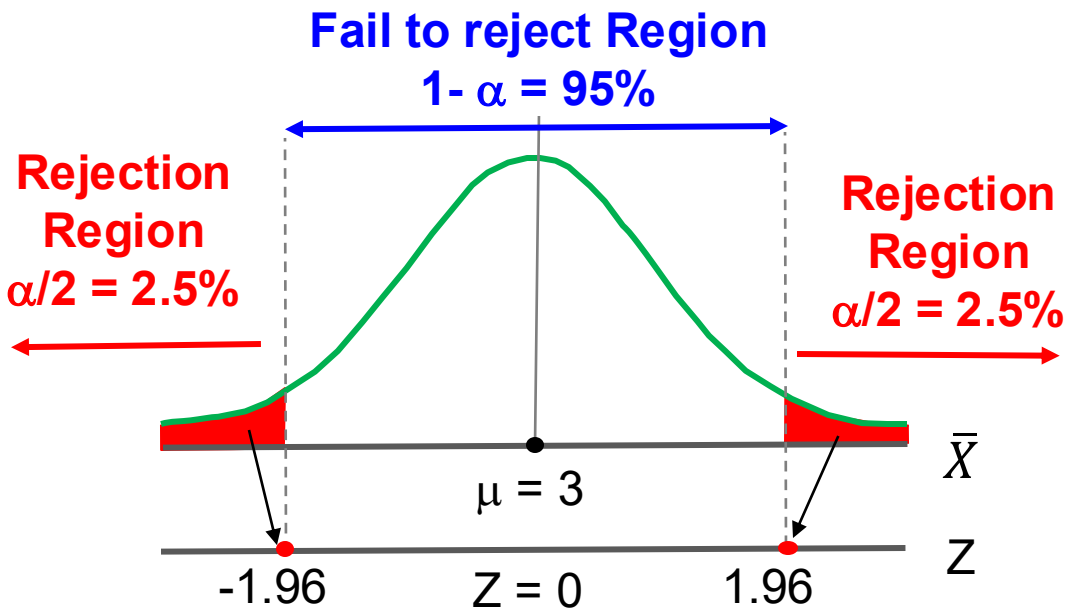
Given :

- Population variance = 1
- Sample Mean = 2.8
- Test for significance level = 5%, conf.lv. = 95%
- Number of Sample = 64

SOLUTION (TRADITIONAL METHOD)

Step-1: State the H_0 and the H_a.

- $H_0 : \mu = 3$
- $H_a : \mu \neq 3$



Step-2: Specify the significance level.

$\alpha = 5\% = 0.05$

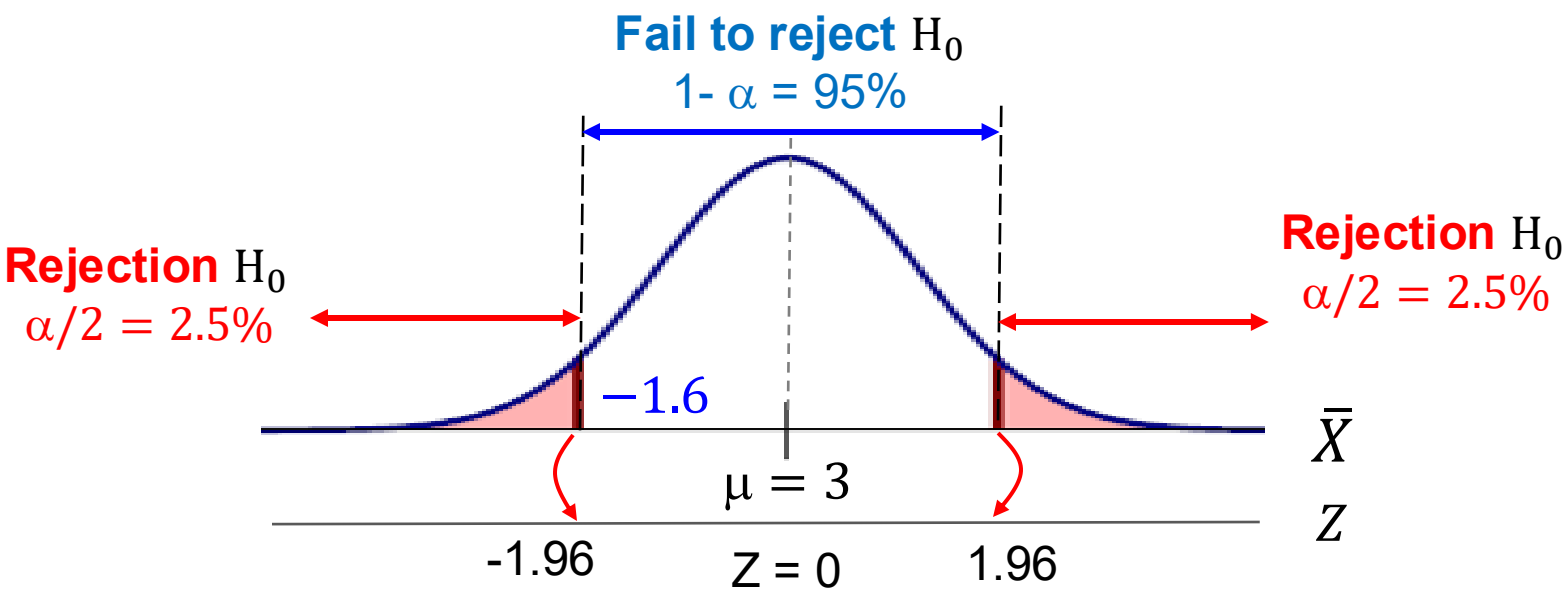
➔ $Z_{\alpha/2} = Z_{0.025} = -1.96$

SOLUTION (TRADITION METHOD)

Step-3 : Determine the test statistic.

We use Z-distribution since σ is known and population being sampled is normally distributed.

$$\bar{X} = 2.8 \rightarrow Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{2.8 - 3}{1/8} = -1.6$$



Step-4: Formulate the decision rule.

Reject H_0 *if* $|Z| > Z_{\alpha/2}$

$$-1.96 < Z(= -1.6) < 1.96$$

➔ *Fail to reject* H_0

Step-5: Make decision and interpret the result.

Z-statistics of –1.6 does not fall into the rejection region, H_0 is not rejected. We conclude that the population mean is not different from 3

So, we would report that the student watch an average of 3 hours social media per day is true.

P-VALUE APPROACH

- **P-value**: Probability of obtaining a test statistic **equal to or more extreme** than the observed sample value **given the H_0 is true**
  - The P-value is also called the observed level of significance
  - It is the smallest value of α for which H_0 can be rejected

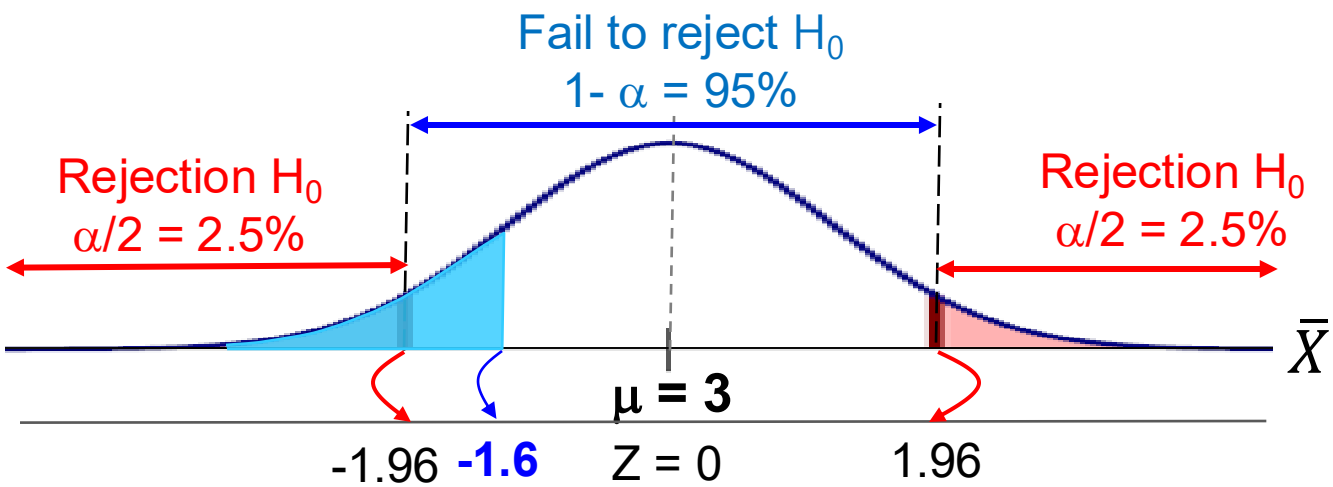
- **Compare the p-value with α**
  - If p-value < α, reject H_0
  - If p-value $\geq \alpha$, do not reject H_0

SOLUTION (P-VALUE)

From previous example: $H_0 : \mu=3$, $H_a : \mu \neq 3$

$\bar{X} = 2.8 \rightarrow Z_{statistic} = -1.6,$ $P[Z \leq -1.6] = 0.05480$

P-Value (for two-side) = 2 x 0.05480 = 0.1096



P-Value(0.1096) > α(0.05) results in Fail to reject H_0

HYPOTHESIS TESTING FOR A POPULATION MEAN WITH KNOWN σ – EX#2

Q: Is the temperature required to damage a computer on the average less than 110 degrees?

Experiment that the 20 computers were tested to see what minimum temperature will damage the computer. We found that the damaging temperature averaged 109 degrees
with known $\sigma = 3$

What is the answer if we decide to test the computer temp. with significant level = 0.05

SOLUTION

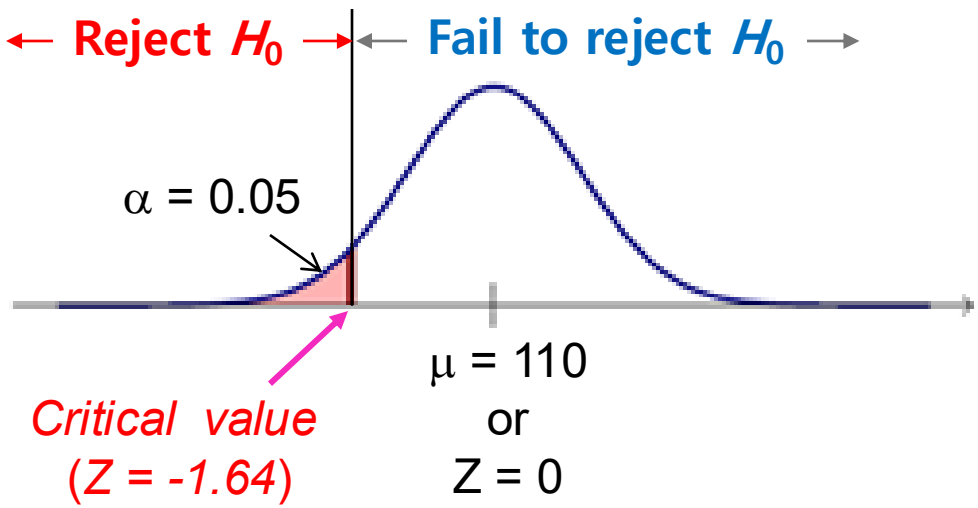
- define alternative hypothesis and null hypothesis

$$H_0 : \mu \geq 110$$

$$H_a : \mu < 110$$

- The criteria for a decision => one-tailed test

$\alpha = 0.05$

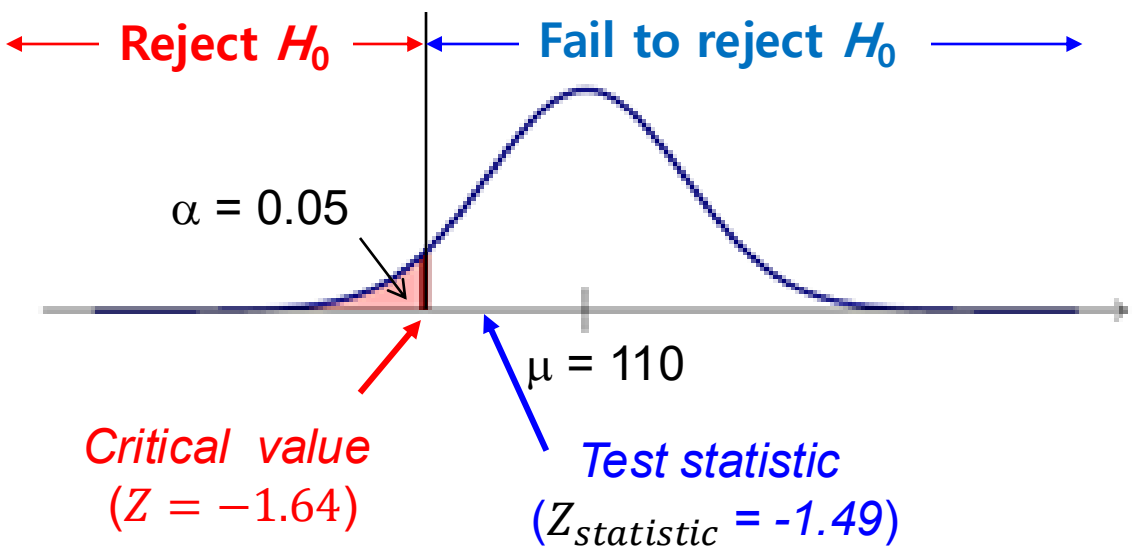


SOLUTION (CONT.)

- Calculate the test statistic.

$$\sigma = 3,\ \bar{X} = 109,\ \ \mu = 110$$

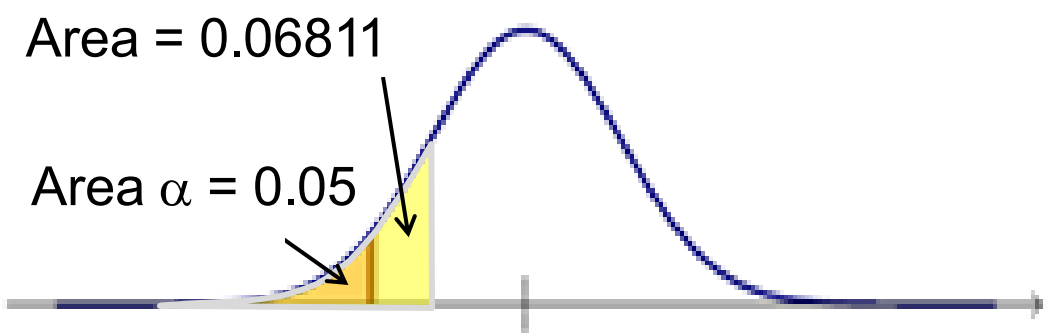
$$Z_{statistic} = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{109 - 110}{3/\sqrt{20}} = -1.49$$



SOLUTION (CONT.)

- Compute the P-value.
  - from the Z-table at $Z = -1.49$,

$$\mathrm{P}(Z \leq -1.49) = 0.06811$$



MAKE A DECISION

- **Traditional Method** **(Critical value as the test statistic)**
  - Test statistic does not fall within the critical region ➔ ***fail to reject H_0***
- **P-value**
  - P-value (0.06811)> α(0.05) ➔ ***fail to reject H_0***

Conclusion: We fail to reject the null hypothesis and conclude that *there is insufficient evidence to suggest that the temperature required to damage a computer on the average less than 110 degrees.*

TESTING FOR THE POPULATION MEAN : UNKNOWN σ

- When the population std.dev (σ) is unknown, the sample standard deviation (S) is preferred.
- The **t-distribution is used as test statistic**

$$t = \frac{\bar{X} - \mu_0}{\frac{S}{\sqrt{n}}}$$

Using t-distribution with $n-1$ degrees of freedom, where :

$\bar{X}$: the sample mean.

μ_0 : the hypothesized population mean.

S : the sample standard deviation.

n : the number of observations in the sample

TESTS CONCERNING SAMPLE MEAN (VARIANCE UNKNOWN)

A hard disk manufacturer introduces a new method of assembling a particular component. The old method had a mean assembly time of 42 minutes. The manufacturer would like the assembly time to be as short as possible, and so he expects the new method to have a smaller mean.

A random sample of assembly times in minutes taken after the new method had become established was

27	39	28	41	47	42	35	32	38

At α= 5%, is there an evidence that the new method works?

TESTS CONCERNING SAMPLE MEAN (VARIANCE UNKNOWN) CONTINUED

Step-1: State the H_0 and the H_a

$$H_o: \mu \geq 42, \ H_a: \mu < 42$$

Step-2: significance level α = *0.05* → $(t = -1.860)$

Step-3: Calculate the test statistic

$$\bar{X} = \frac{27+39+28+41+47+42+35+32+38}{9} = 36.56$$

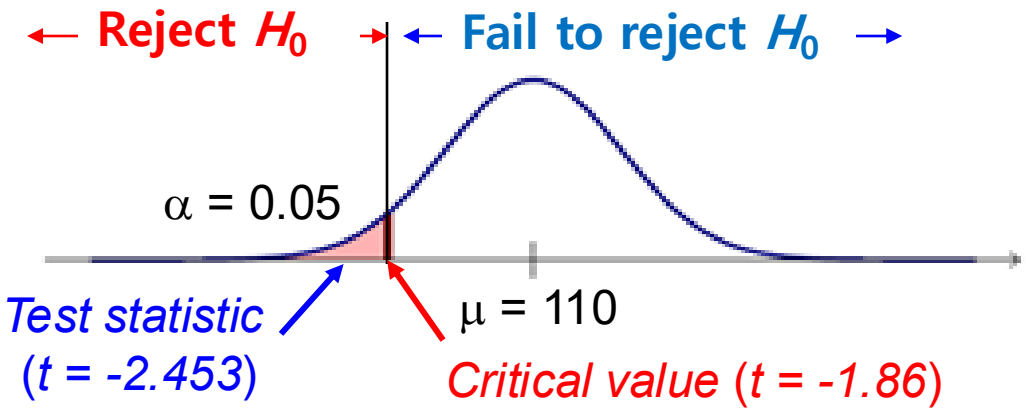
$$S = \sqrt{\frac{\sum(x-\bar{X})^2}{n-1}} = 6.654$$

$$t_{statistic} = \frac{36.56-42}{6.654/\sqrt{9}} = -2.453$$ (test statistic)

TESTS CONCERNING SAMPLE MEAN (VARIANCE UNKNOWN) -CONTD

Step-4: Formulate the decision rule = left-tailed test

Reject H_0 if $t < -t_{\alpha/2,n-1}$



Step-5: Make a decision

The test statistic falls in the rejection region, then reject H_0. $(t_{statistic} < t_{critical})$

TESTING FOR A POPULATION MEAN WITH AN UNKNOWN σ EXAMPLE

The current rate for producing 5 A fuse at a Co. is 250 piece per hour. A new machine has been purchased and installed that, according to the supplier, will increase the production rate.

A sample of 10 randomly selected hours from last month revealed the mean hourly production on the new machine was 256 units, with a sample standard deviation of 5 per hour.

At the 0.05 significance level can the Co. conclude that the new machine is faster?

TESTING FOR A POPULATION MEAN WITH AN UNKNOWN σ EXAMPLE CONTINUED

Step 1: State the null and the alternate hypothesis

$$\mathrm{H_0}: \mu \leq 250; \qquad \mathrm{H_a}: \mu > 250$$

Step 2: level of significance α = 0.05, unknown σ

degrees of freedom for mean = 10 – 1 = 9

$$t_{\alpha,9} = 1.833$$

Step 3 : Find a test statistic, unknown σ

then use the t-distribution and n is less than 30.

$$t_{statistic} = \frac{\bar{X} - \mu_0}{\frac{S}{\sqrt{n}}} = \frac{256 - 250}{\frac{6}{\sqrt{10}}} = 3.162$$

t Table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$
one-tail	**0.50**	**0.25**	**0.20**	**0.15**	**0.10**	**0.05**	**0.025**	**0.01**
two-tails	**1.00**	**0.50**	**0.40**	**0.30**	**0.20**	**0.10**	**0.05**	**0.02**
df								
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764

TESTING FOR A POPULATION MEAN WITH AN UNKNOWN σ EXAMPLE CONTINUED

Step 4: State the decision rule.

$$t_{statistic}(3.162) > t_{critical}(1.833)$$

Then Reject H_0

Step 5: Make a decision

The null hypothesis is rejected. That is the mean number produced is more than 250 per hours.

TESTS CONCERNING A PROPORTION

- The sample proportion is denoted by $\hat{p}$
- The test statistic is computed as follows:

$$Z_{statistic} = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}}$$

Where:

p_0 : population proportion

$\hat{p}$: sample proportion

n : sample size

ASSUMPTIONS IN TESTING A POPULATION PROPORTION USING THE Z-DISTRIBUTION

- A random sample is chosen from the population. Sample proportion in the category of interest denoted by $\hat{p}$

$$\hat{p} = \frac{X}{n} = \frac{\text{number of interest in sample}}{\text{sample size}}$$

When both $\boldsymbol{np}$ and $\boldsymbol{n(1-p)}$ are at least 5, $\hat{p}$ can be approximated by a normal distribution

TEST STATISTIC FOR TESTING A SINGLE POPULATION PROPORTION - EXAMPLE

A marketing company claims that it receives 8% responses from its ads. To test this claim, a random sample of 500 were surveyed with 25 responses. Test at the α = 0.05 significance level.

- Check: $\hat{p}$ is in a normal distribution ?
  - $np_0 = 500(0.08) = 40$, *more than 5*
  - $n(1-p_0) = 500(0.92) = 460$, *more than 5*
- Sample proportion $\hat{p} = \frac{x}{n} = \frac{25}{500} = 0.05$

TEST STATISTIC FOR TESTING A SINGLE POPULATION PROPORTION - EXAMPLE

Step-1: State $\mathrm{H_0}$ and $\mathrm{H_a}$

$\mathrm{H_0}: p = 0.8;\ \ \mathrm{H_a}: p \neq 0.8$

Step 2: level of significance. α = 0.01

Apply Z-distribution since np and $n(1 - p_0) \geq 5$

$$Z_{\alpha/2} = -1.96$$

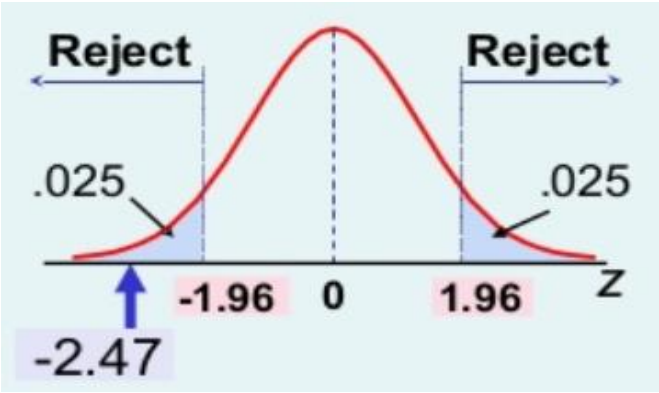
Step 3: Calculate the test statistic

$$Z_{statistic} = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} = \frac{0.05 - 0.08}{\sqrt{\frac{0.08(1-0.08)}{500}}} = -2.47$$

Step 4: set the decision rule

$$Z_{statistic}\,(-2.47) < Z_{critical}(-1.96)$$

Reject H_0 at $\alpha = 0.05$



Step 5: Make a decision

There is sufficient evidence to reject the company's claim of 8% response rate.

MORE EXAMPLE 1

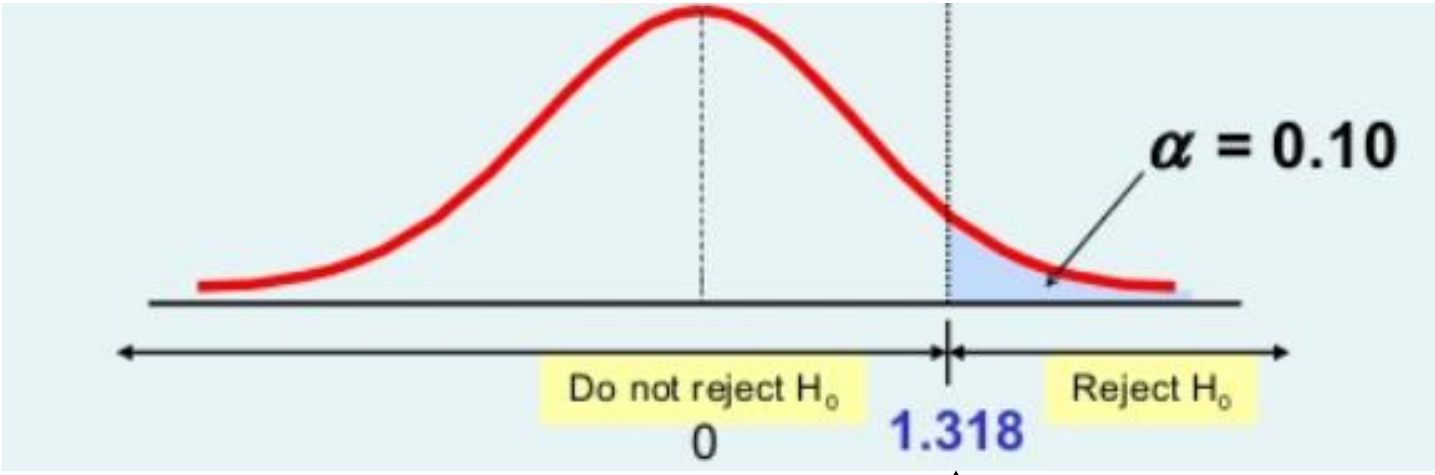
A company manager thinks that customer monthly subscription have increased, and now average over \$52 per month.

The company wishes to test this claim. Suppose that significant is 0.10, with given $n = 25,\ \bar{X} = 53.1,$ $S = 10.$

- $H_0 : \mu \leq 52$
- $H_a : \mu > 52$

EXAMPLE : FIND REJECTION REGION

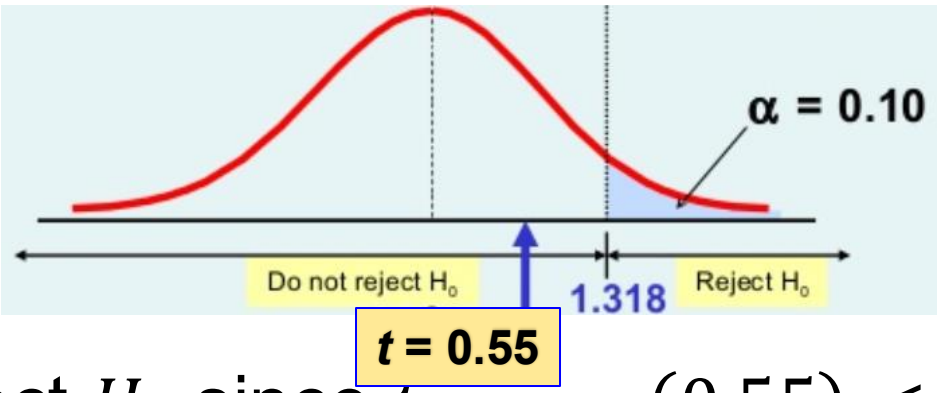
- *Significant*, $\alpha = 0.10$, $t_{\alpha,24} = 1.318$



Reject H_0 if $t > 1.318$

EXAMPLE: TEST STATISTIC

$$t_{statistic} = \frac{\bar{X} - \mu_0}{\frac{s}{\sqrt{n}}} = \frac{53.1 - 52}{\frac{10}{\sqrt{25}}} = 0.55$$



- Fail to reject H_0 since $t_{statistic}(0.55) < t_{\alpha,24}(1.318)$
- There is not sufficient evidence that the mean is over $52

MORE EXAMPLE 2

A batch of 100 resistors have an average of 102 Ohms. Assuming a population standard deviation of 8 Ohms.

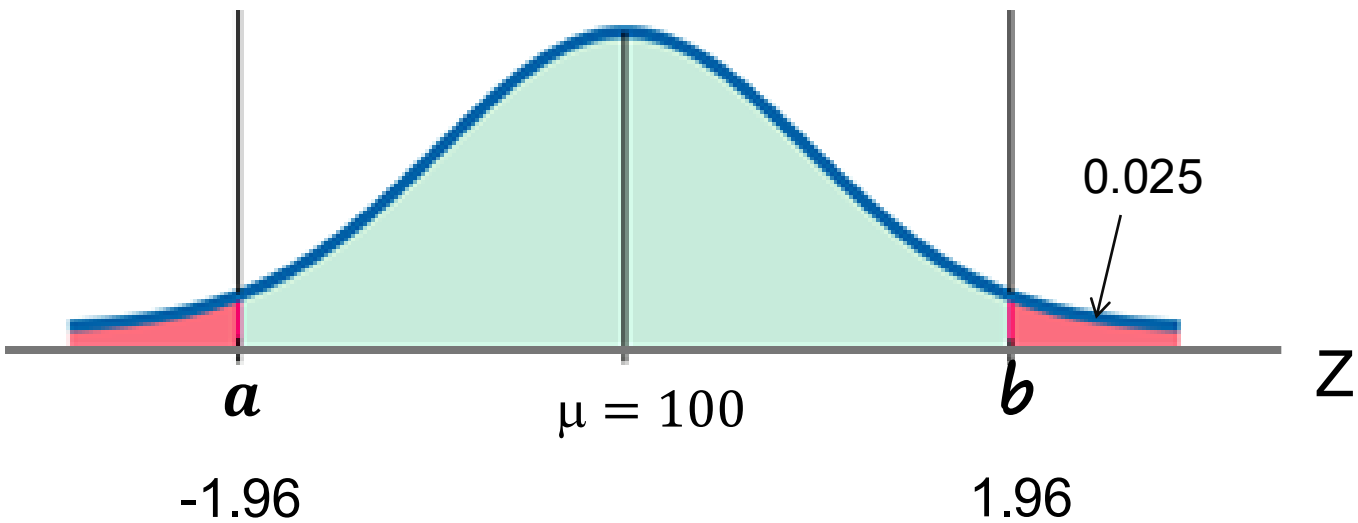
Test whether the population mean is 100 Ohms at a significance level of $\alpha = 0.05$.

SOLUTION

$H_0 : \mu = 100$

$H_a : \mu \neq 100$

$\alpha = 0.05$



SOLUTION (CONT.)

Traditional method (critical value)

The test statistic is specified by the problem to be the sample mean $\bar{X}$.

Reject H_0 if $X < a$ or $X > b$ with

$$a = \mu_0 - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} = \mu_0 - z_{0.025}\frac{\sigma}{\sqrt{100}} = 100 - 1.96(\frac{8}{10}) = 98.432$$

$$b = \mu_0 + z_{\alpha/2}\frac{\sigma}{\sqrt{n}} = 100 + 1.96(\frac{8}{10}) = 101.568$$

Decision : The test statistic on a sample is $\bar{X} = 102 > b$. Therefore, reject H_0

SOLUTION (CONT.)

P-Value

$$z = \frac{102 - 100}{8/\sqrt{100}} = 2.5$$

$P(Z < -2.5) = 0.00621$

P-value $= 2 \times 0.00621 = 0.01242$

Decision : The *P*-value of 0.01242 is less than the significance level (α = 0.05), so we reject the null hypothesis.

MORE EXAMPLE 3

A quality control engineer finds that a sample of 100 light bulbs had an average life-time of 470 hours.

- Assuming a population standard deviation of σ = 25 hours, test whether the population mean is 480 hours at a significance level of α = 0.05.

SOLUTION

$H_0 : \mu \geq 480$

$H_a : \mu < 480$

$\alpha = 0.05$

The test statistic is the sample mean $\bar{X}$.

Reject H_0 if $\bar{X} < \mu_0 - z_\alpha \frac{\sigma}{\sqrt{n}}$

$$\mu_0 - z_\alpha \frac{\sigma}{\sqrt{n}} = 480 - 1.645 \times \frac{25}{10} = 475.9$$

Since $\bar{X} = 470 < 475.9$, we reject H_0.

EXAMPLE

In communication system, A transmission time of a random sample of 12 packets are as follows:

21 18 19 16 18 24 22 19 24 14 18 15

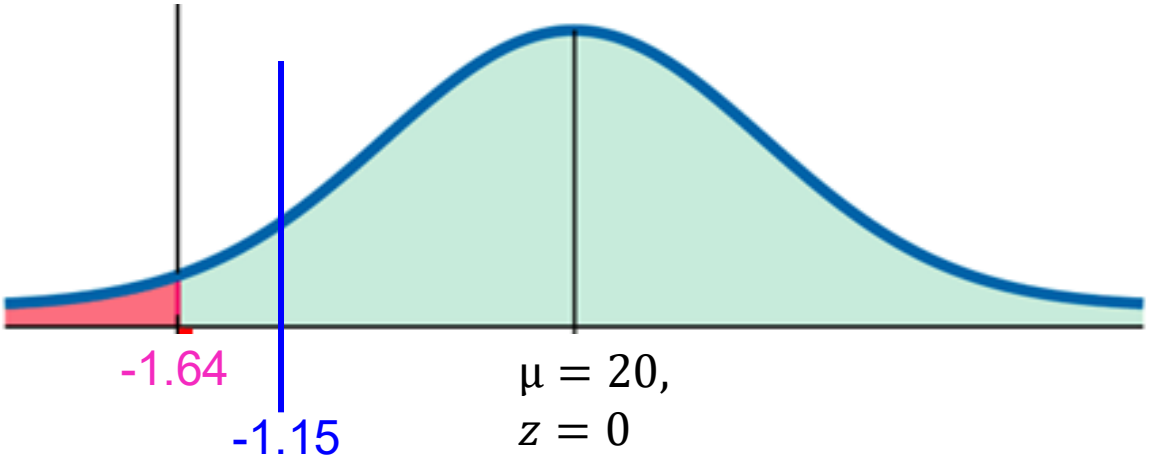
The transmitter claims that the mean of transmission time is less than 20. Assuming a transmission time is to be normally distribution with standard deviation of 3.

With significance level is 0.05, carry out an appropriate hypothesis test to advise as the validity of a transmitter claim.

SOLUTION

We asking this is true or not?

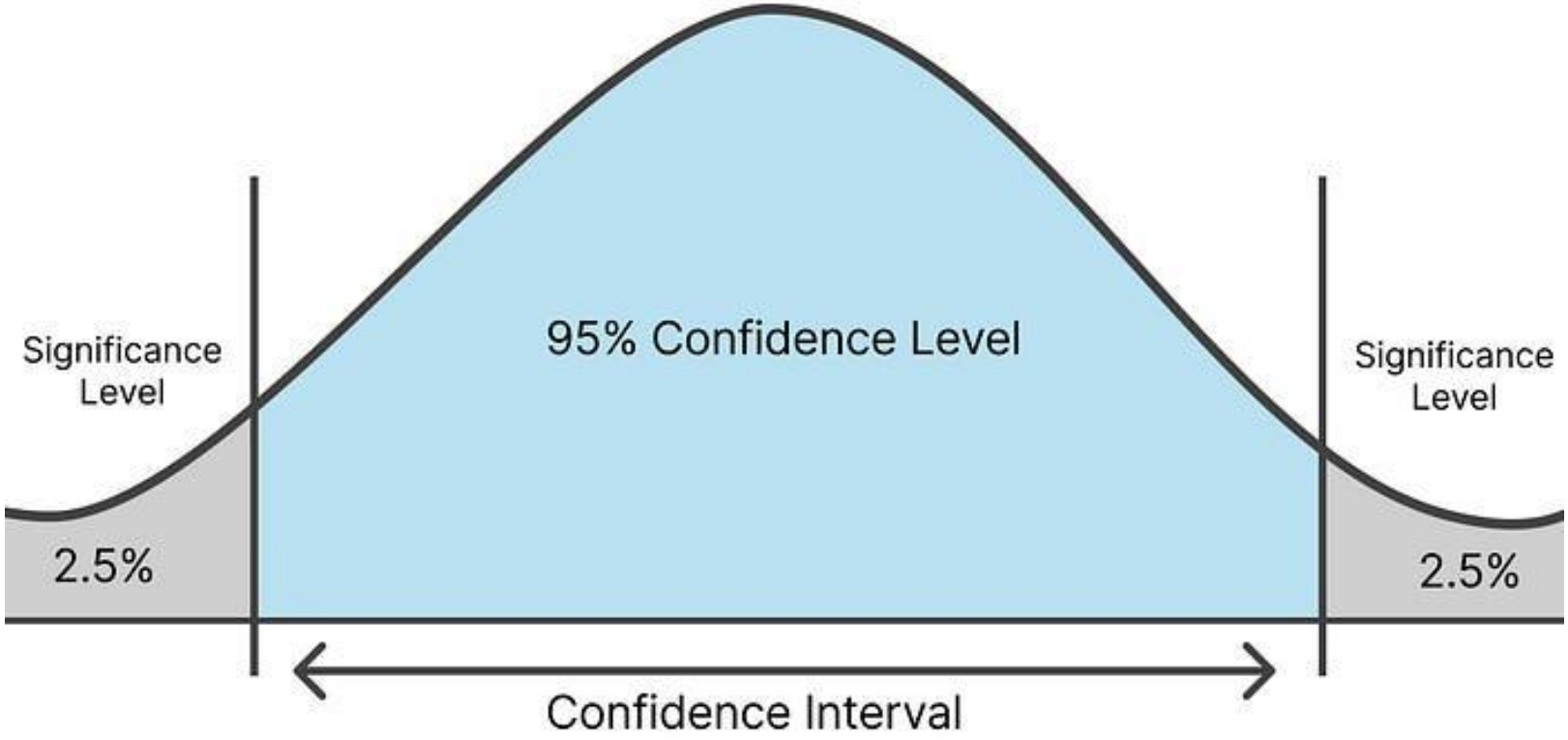
- $\mathrm{H}_0 : \mu \geq 20,\ \mathrm{Ha} : \mu < 20$, test at 20
- α = 0.05 that is $P[Z < -1.64] = 0.05$
- Reject H_0 when $Z_{statistic}$ fall within (–inf ,-1.64)



- $\bar{X} = 19,\ Z = \frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}} = \frac{19-20}{3/\sqrt{12}} = -1.15$

- $Z_{statistic} > Z_{\alpha}$ ➔ Fail to reject H_0

- Thus, there is no evidence, at the 5% level of significance, to support the transmitter's claim.
- The tx time of comm system is not less than 20 at 5% level of significant.


Significance Level
95% Confidence Level
Significance Level
2.5%
2.5%
Confidence Interval

MORE EXAMPLE

We have a sample of 50 response time of computers having a mean of 98 ns. Assume that the sample is a simple random sample and that the population standard deviation σ is known to be 20 ns.

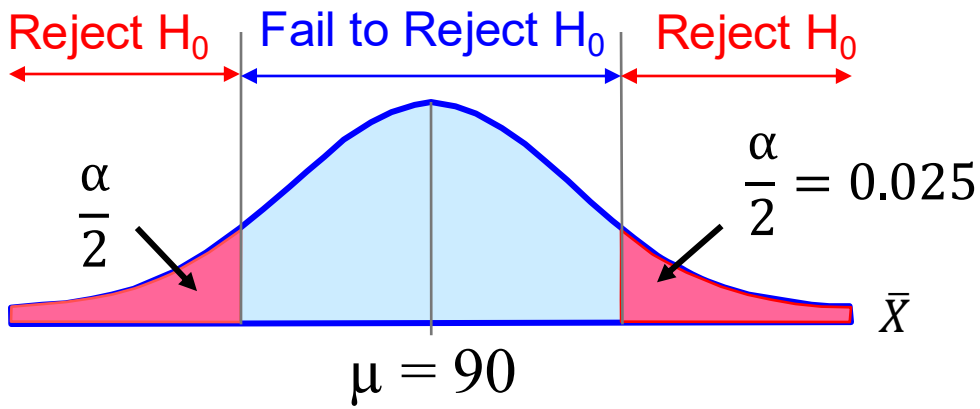
Test the common belief that the mean response time of good computers is equal to 90 ns. with 0.05 significance level

- Use the *P*-value method.
- Use the traditional method

SOLUTION (P-VALUE)

H_0: $\mu = 90$, H_a : $\mu \neq 90$ *becomes a two-tailed test*

$\alpha = 0.05$, $\bar{X} = 98$, $\sigma = 20$, n = 50

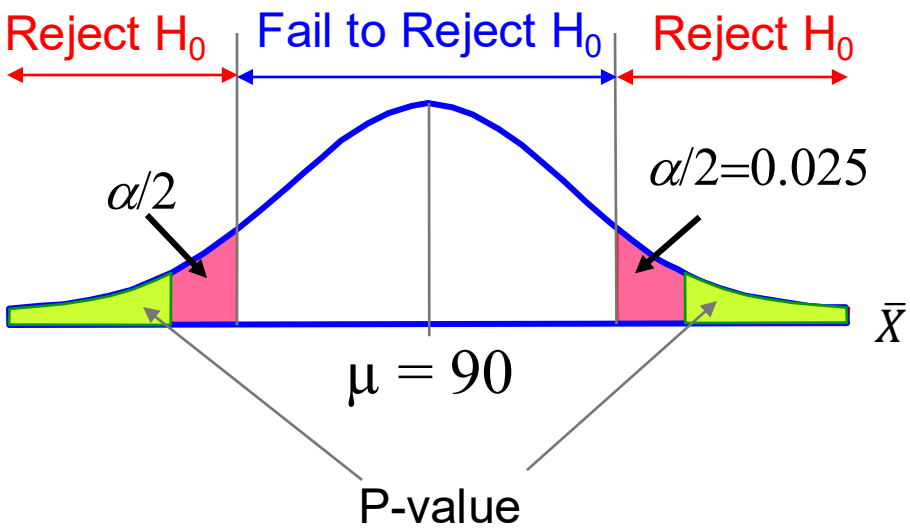


Compute the value of the test statistic.

$$Z_{statistic} = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{98 - 90}{20/\sqrt{50}} = 2.83$$

Computer the p-value.

$$P(Z \leq 2.83) = 0.998$$



This notice that if z-statistic > 0, then p-value = 1-p(z)

Two-tailed test

$$P_value = 2(1 - 0.998) = 2 \times 0.002 = 0.004$$

Make conclusion about H_o

- Because *p-value* = 0.004 $\leq$ 0.05 ➔ Reject H_o

Good computer is not meeting the response time of 90 ns

SOLUTION(USE CRITICAL VALUE)

Critical Value Approach

- Determine the critical value and rejection rule
  - $\alpha = 0.05 \rightarrow Z_{0.025} = -1.96, \quad Z_{1-0.025} = 1.96$
  - Reject H_0 if $Z_{statistic} \geq 1.96$
- Make conclusion about H_0
  - $Z_{statistic} = 2.83 >$ critical z = 1.96 ➔ reject H_0

There is sufficient evidence to conclude that the mean response time of good computers differs form 90 ns

SOLUTION (CONFIDENCE INTERVAL)

$$P\left[\bar{X} - Z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right] = 1 - \alpha$$

$$P\left[98 - 1.96\frac{20}{\sqrt{50}} \leq \mu \leq 98 + 1.96\frac{20}{\sqrt{50}}\right] = 0.95$$

$$P[92.45 < \mu < 103.54] = 0.95$$

We are 95% confident that the limits of 92.45 and 103.54 contain the true value of μ, so it appears that 90 ns cannot be the true value of μ

MORE EXAMPLE

- The lifetime of a CPU is known to be normally distributed with a standard deviation of 2.
- A sample of 6 CPUs yielded the following observations 2.0, 1.3, 6.0, 1.9, 5.1, 4.
- At α = 0.05, is there the reason to believe that the mean life of the CPU is greater than 2? And Find 95% confident interval

- Is there reason to believe that the mean life of the CPU is greater than 2?

$$H_0: \mu \leq 2; \quad H_a: \mu > 2$$

$$\bar{X} = \frac{2 + 1.3 + 6 + 1.9 + 5.1 + 4}{6} = 3.383$$

$$\alpha = 0.05 \rightarrow Z_{0.95} = 1.64$$

$$Z_{statistic} = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{3.383 - 2}{2/\sqrt{6}} = 1.6939$$

$$Z_\alpha < Z_{statistic} \rightarrow \text{reject } H_0$$

SOLUTION

- Find a 95% confidence interval

$\bar{X} = 3.383$

95% Confident Interval → $Z_{0.025,\,0.975} = \pm 1.96$

$\mu = \bar{X} \pm Z_{0.0250.975} \frac{\sigma}{\sqrt{n}}$

$= 3.383 \pm 1.96 \left(\frac{2}{\sqrt{6}}\right)$

$= 3.3383 \pm 1.6003$

$$\mu = (1.738, 4.9003)$$

CONNECTION BETWEEN TWO TAIL TESTS AND CONFIDENCE

Consider a production line of resistors that are supposed to be 100 Ohms.

Assume σ = 8, n = 64 and $\bar{X} = 98$, the 95%CI is :

$$98 - 1.96\frac{0.8}{\sqrt{64}} \leq \mu \leq 98 + 1.96\frac{0.8}{\sqrt{64}}$$

$$97.804 \leq \mu \leq 98.196$$

Since this interval does not contain the hypothesized mean(100), so we reject the null hypothesis at α = 5%

ERRORS IN HYPOTHESIS TEST

- **Type I Error**
  - Reject a true null hypothesis (false-positive)
  - The probability of a Type I Error is α
    - ✓ *Type I Error* is a level of significance of the test
    - ✓ Set by researcher in advance

- **Type II Error**
  - Failure to reject false null hypothesis (false-negative)
  - The probability of a Type II Error is β

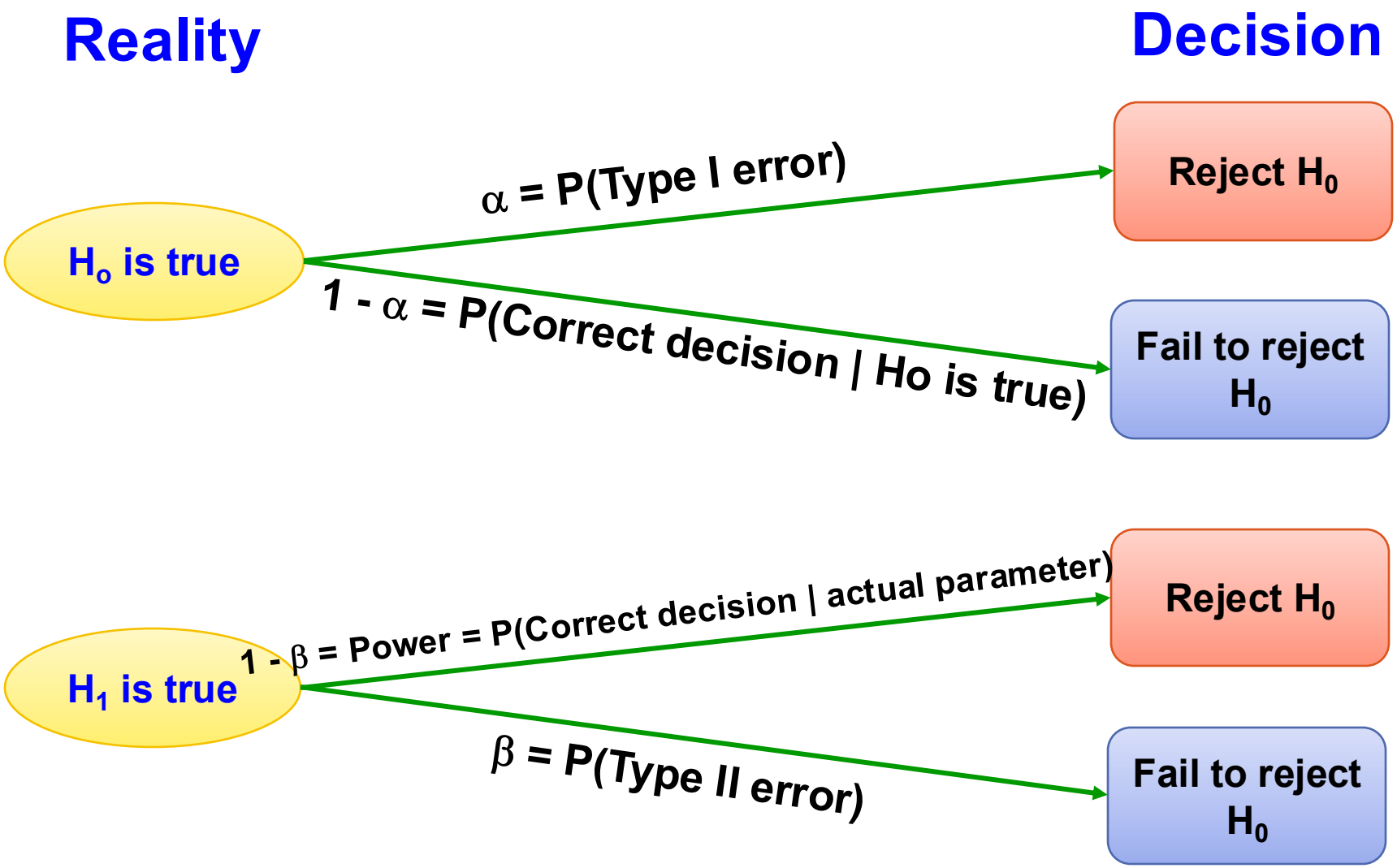
TYPE I AND TYPE II ERRORS

Decision \ Population Condition	H_0 is True (positive)	H_0 is False (negative)
Not reject H_0 (Retain $\mu = \mu_0$)	Correct Decision ($1-\alpha$)	**Type II Error (β)**
Reject H_0 (Conclude $\mu > \mu_0$)	**Type I Error (α)**	Correct Decision ($1-\beta$)

Type I Error = P(Reject H_0 | H_0 is true)

Type II Error = P(Fail to reject H_0 | H_1 is true)

REALITY TO DECISION



POSSIBLE ERRORS IN HYPOTHESIS TEST DECISION MAKING

- The confidence coefficient (1-α) is the probability of not rejecting H_0 when it is true.
  - The confidence level of a hypothesis test is (1- α)*100%

- The power of a statistical test (1-β) is the probability of rejecting H_0 when H_0 is false.

TYPE I & II ERROR RELATIONSHIP

- Type-I and Type-II errors cannot happen at the same time
  - A Type-I error can only occur if **H_0 is true**
  - A Type-II error can only occur if **H_0 is false**

- If Type-I error probability (α) increases, then Type-II error probability (β) decreases

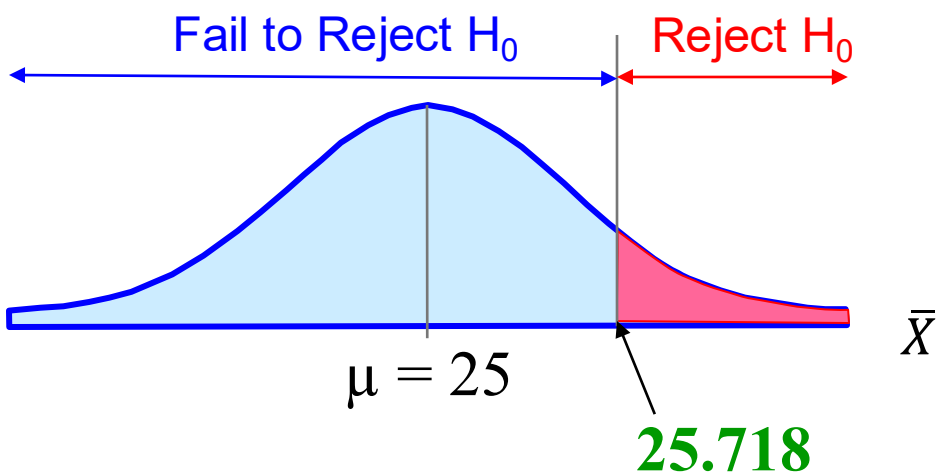
EXAMPLE : TYPE I, II ERROR

- Consider the following hypothesis test

$$H_0: \mu \leq 25$$

$$H_a: \mu > 25$$

Assume $\bar{X}_{critical} = 25.718, \sigma = 2.4, \ n = 30$



EXAMPLE TYPE-I ERROR

- **Find Type-I Error**

P(Type-I Error) = P(reject H_0 | H_0 is true)

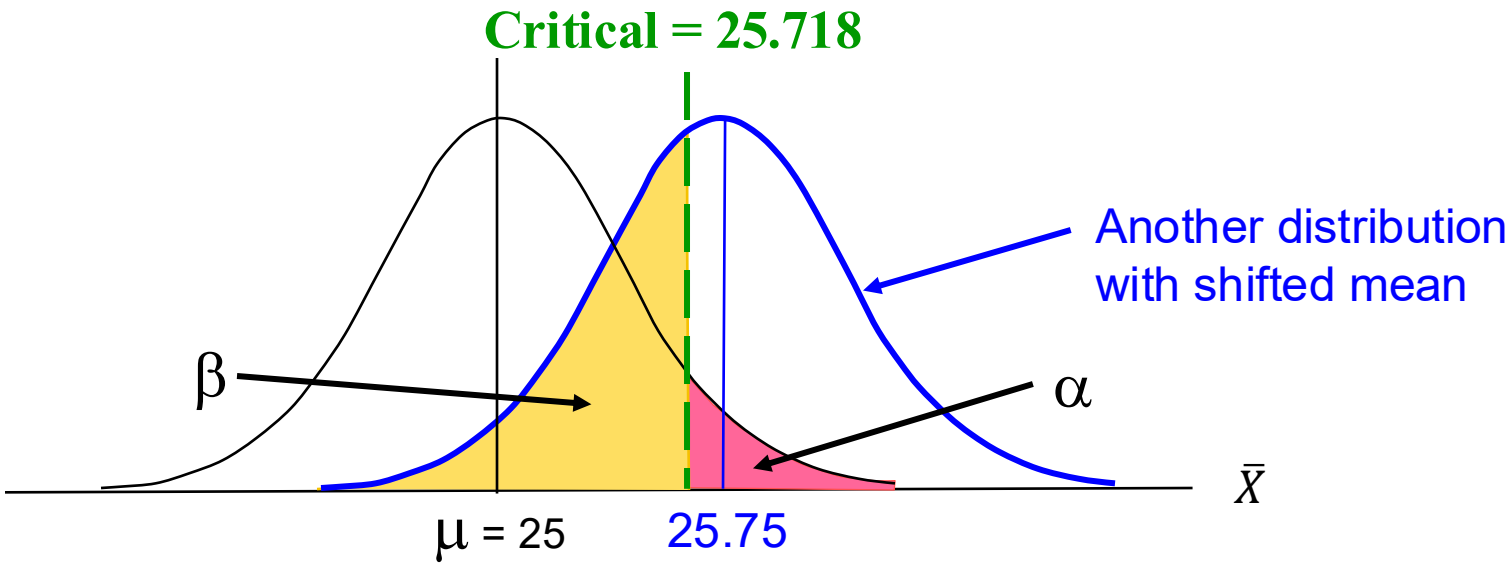
$$= P(\bar{X} > 25.718 \mid \mu = 25)$$

$$= P\left(\bar{Z} > \frac{25.718-25}{2.4/\sqrt{30}}\right)$$

$$= P(Z > 1.64) = 0.05$$

EXAMPLE-TYPE II ERROR

- Find Type-II error when true mean = 25.75



P(Type-II error | μ = 25.75)

= P(fail to reject H_0 | H_a is true)

EXAMPLE-TYPE II ERROR

P(Type-II Error | μ = 25.75)

= P(fail to reject H_0 | H_1 is true)

$$= P(\bar{X} < 25.718 \mid \mu = 25.75)$$

$$= P\left(\bar{Z} < \frac{25.718 - 25.75}{2.4/\sqrt{30}}\right)$$

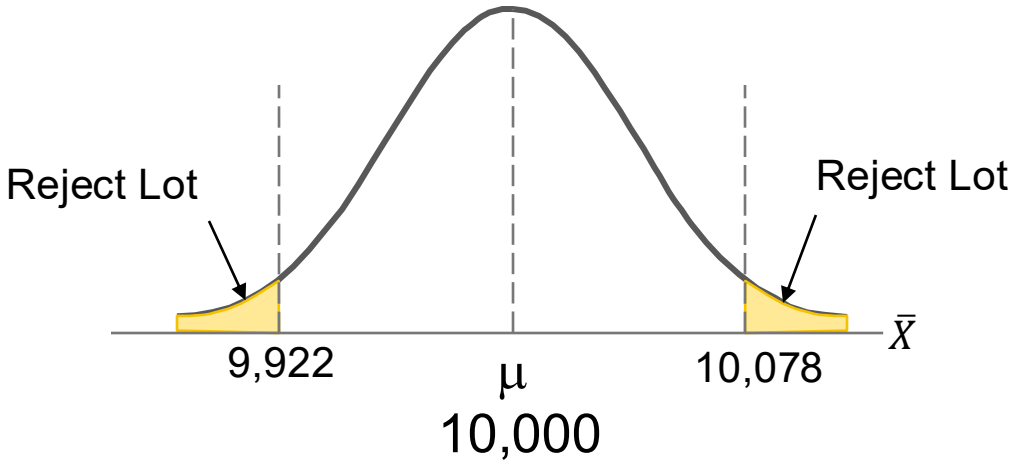
$$= P(Z < 0.07) = 0.4721$$

TYPE II ERROR-EXAMPLE

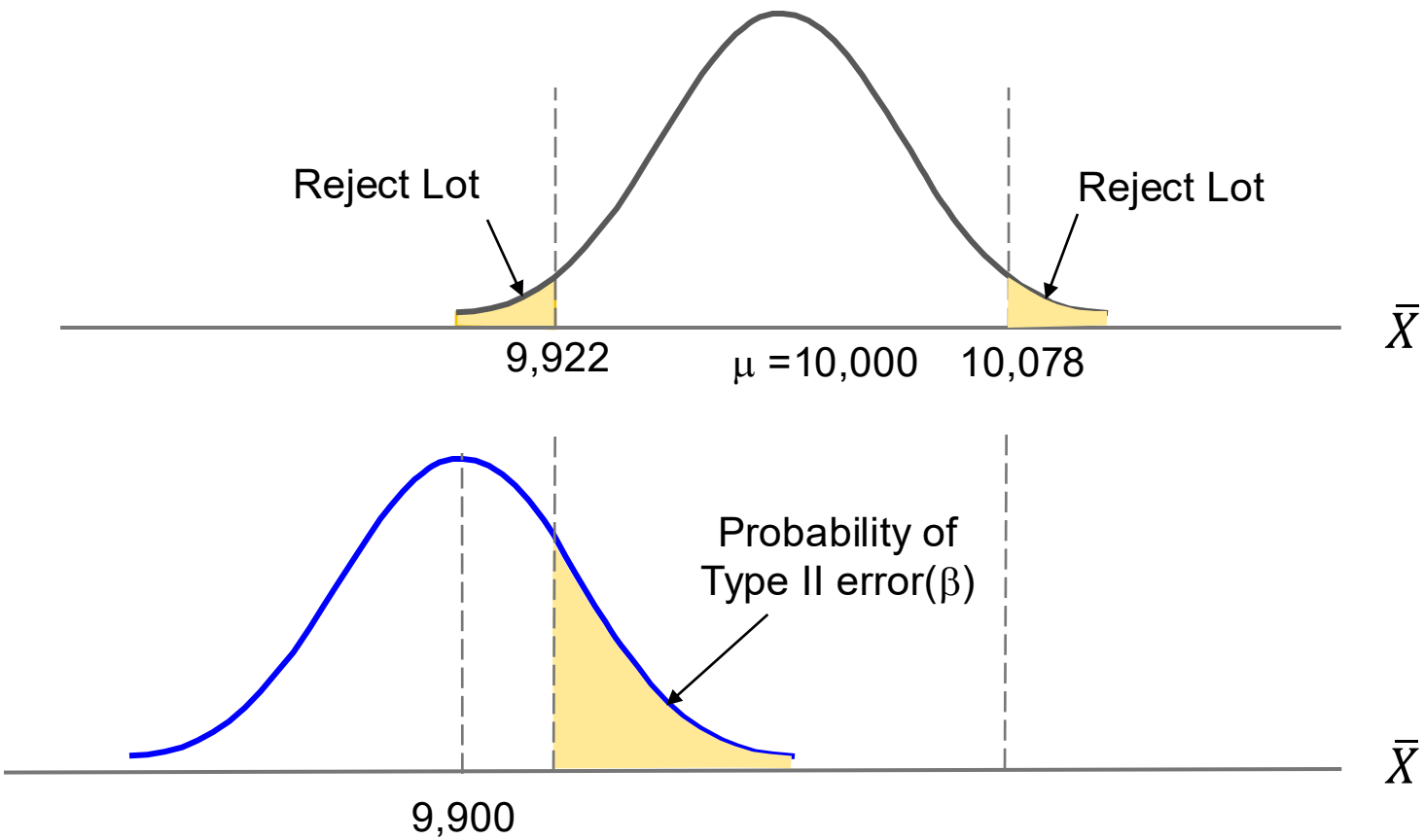
A manufacturer purchases steel bars to make cotter pins. Experience indicates that the mean tensile strength of all incoming shipments is 10,000 psi. σ is 400 psi.

To make a decision about incoming shipments of steel bars, the manufacturer set up a rule for the quality-control inspector: "Take a sample of 100 steel bars. At the .05 significance level if the sample mean strength falls between 9,922 psi and 10,078 psi, accept the lot. Otherwise, the lot is to be rejected."

$$H_0: \mu = 10000, \qquad H_a: \mu \neq 1000$$



- The mean of the incoming lot is 9,900



n = 100, σ = 400

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{9000 - 9900}{400/\sqrt{100}} = \frac{22}{40} = 0.55$$

$$P[Z > 0.55] = 1 - P[Z < 0.55]$$

$$= 1 - 0.70884$$

$$= 0.29116$$

Type II Error = 0.29116

TYPE II ERROR PROBABILITY

Since we don't actually know what the true mean is (or we wouldn't be hypothesizing something else), we cannot know in practice the type II error rate (β). However, it is affected by a number of things, and we can know about these.

We define a new term **POWER**, this is the **probability of NOT making a type II error (= 1-β)**.

POWER

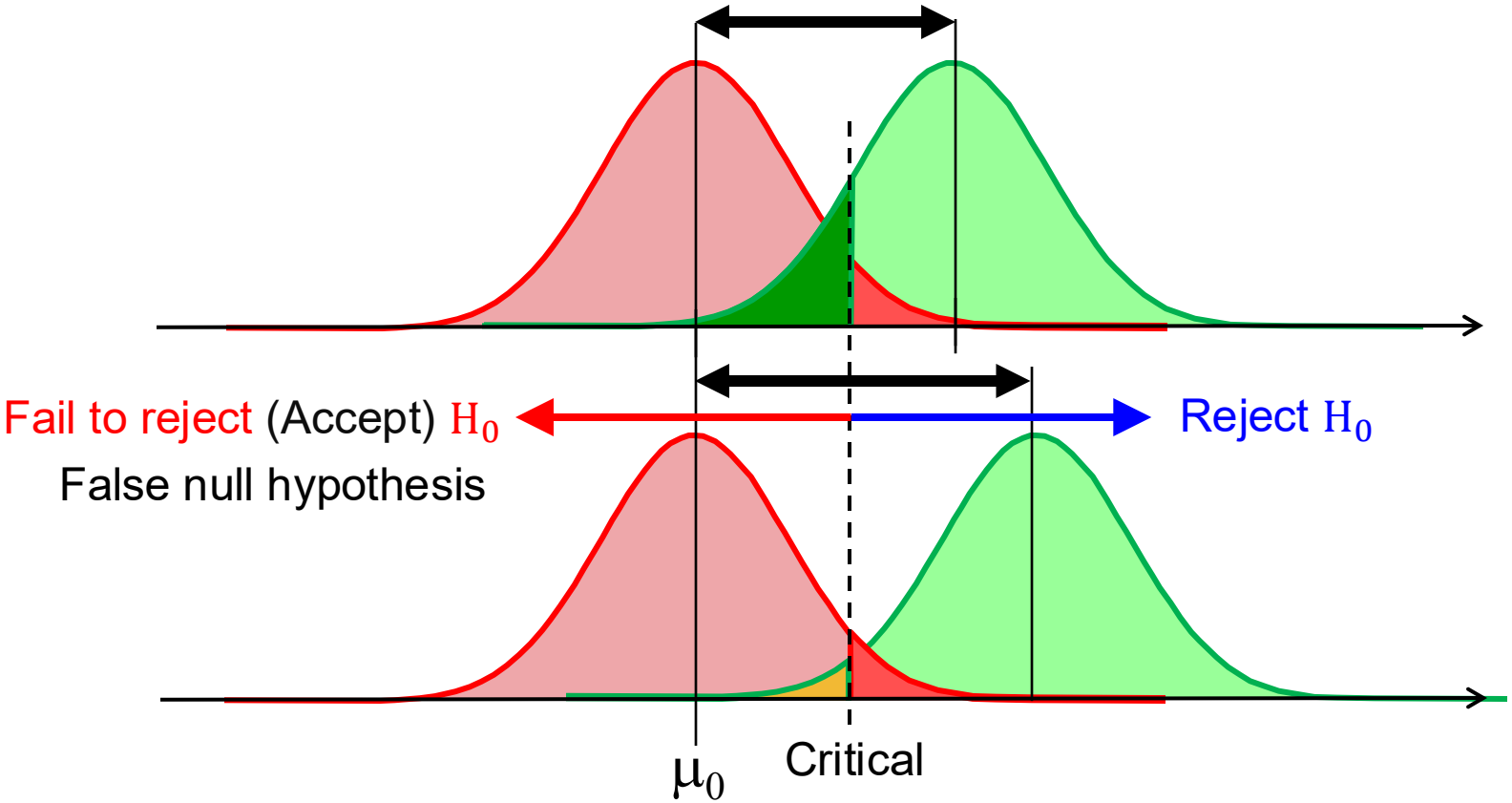
The power of a hypothesis test is the probability (1 - β) of rejecting a false null hypothesis, which is computed by using a particular significance level α and a particular value of the population parameter, that is an alternative to the value assumed true in the null hypothesis.

That is, the power of the hypothesis test is the probability of supporting an alternative hypothesis that is true.

TYPE II ERROR PROBABILITY (CONT.)

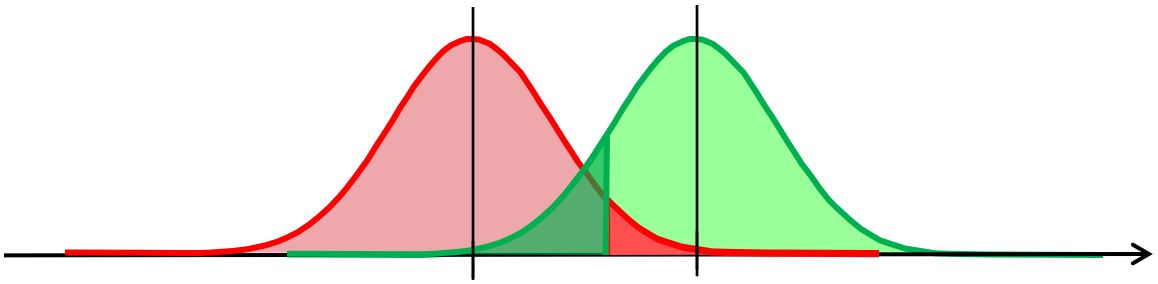
P of rejecting a false null hypothesis

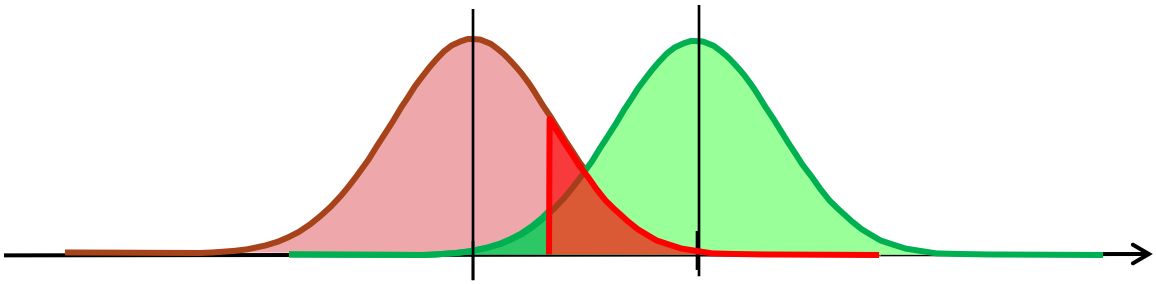
- First, power (1 - β) is affected by the distance between the hypothesized mean (μ_0) and true mean (μ).



TYPE II ERROR PROBABILITY (CONT.)

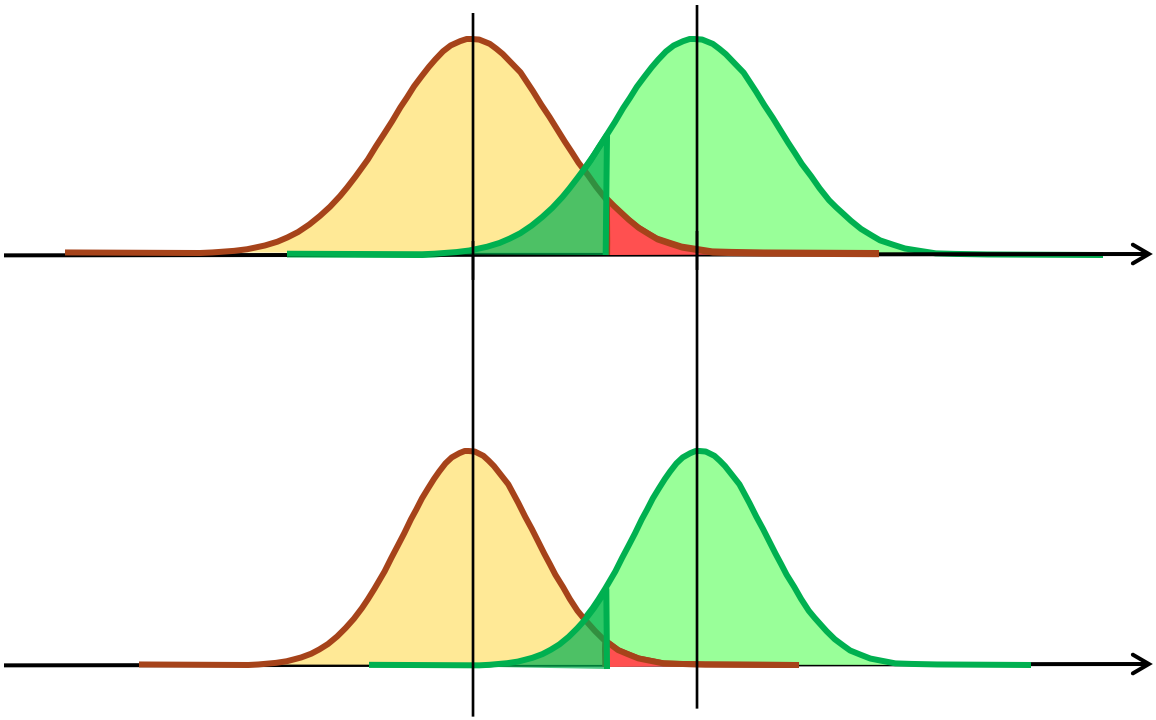
- Second, power is affected by the value chosen for Type I error (α). // critical value depends on α





TYPE II ERROR PROBABILITY (CONT.)

- And third, power is affected by the variability or spread of the distribution. // σ



CONTROLLING POWER

- Which of the 3 factors influencing power can you control?
  - For testing means you may be able to control sample size (n). This reduces the variability and increases power.
  - You probably cannot influence the difference between μ and μ_0.
  - You can choose any value of α. However, this cannot be too small, or Type II error becomes more likely. Too large and Type I error becomes likely.

CONTROLLING TYPE I AND TYPE II ERRORS

- For any fixed α, an increase in the sample size n will cause a decrease in β.
- For any fixed sample size n, a decrease in α will cause an increase in β. Conversely, an increase in α will cause a decrease in β.
- To decrease both α and β, increase the sample size.

T-TABLE

t Table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	**0.50**	**0.25**	**0.20**	**0.15**	**0.10**	**0.05**	**0.025**	**0.01**	**0.005**	**0.001**	**0.0005**
two-tails	**1.00**	**0.50**	**0.40**	**0.30**	**0.20**	**0.10**	**0.05**	**0.02**	**0.01**	**0.002**	**0.001**
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										